How much spike-transmission error can a rate-based motor decoder tolerate?

Much less than a per-spike error model suggests, and it is partly set by temporal integration

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Draft v3, October 2026 Code: https://github.com/youjhun/spike-error-tolerance-decoders

Summary

Question. Lee et al. (2024) showed that an ASBIT wireless link keeps 88.8% (M1) and 85.4% (S1) of wired decoding accuracy at a spike error rate (SER) of 10^{-3}, using an SNN decoder on 1-ms data, and suggested that real populations, which share information, should be even more robust [1]. I asked how much SER classical rate-based decoders tolerate on a public macaque dataset, and why.

Setup. CRCNS pmd-1 (67 M1 + 94 PMd units, 496 trials, 10-ms bins) [2]; Ridge, MLP, LSTM and causal Transformer decoders of 2-D hand velocity from a 200-ms causal window; trial-level 5-fold cross-validation; decoders trained once on clean data and tested on corrupted data. Spike errors were injected under two definitions of SER (Sec. 2).

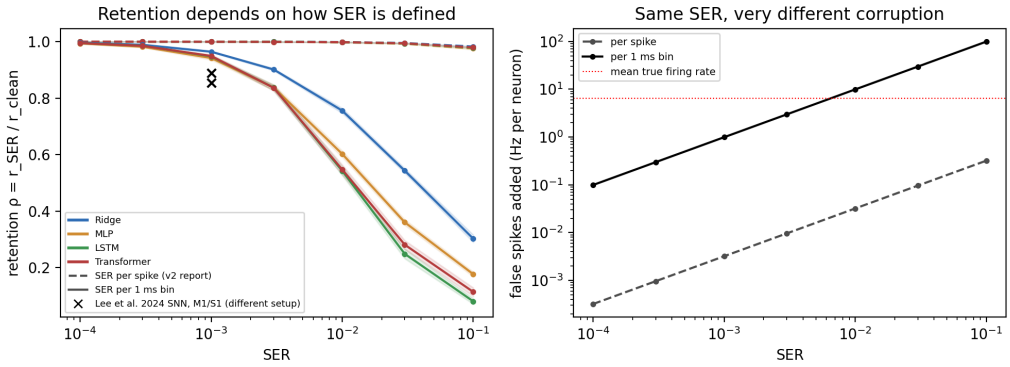


Figure 1: Left: retention $\rho = r_{\mathrm{SER}}/r_{\mathrm{clean}}$ (mean, 95% bootstrap CI) for each decoder. Dashed: SER normalised per spike (used in my previous report). Solid: SER as the fraction of 1-ms bins in error (my reading of Lee et al.). Crosses: Lee et al.'s SNN at SER 10^{-3} (different data and decoder; shown for scale only). **Right:** false spikes each definition adds per neuron, against the mean true rate (6.41 Hz).

Results.

1. **Two SER definitions.** At SER $= 10^{-2}$, Ridge retains $\rho = 0.998$ when errors are counted per spike, but 0.756 [0.747, 0.764] when they are counted per 1-ms bin. The per-bin definition adds 9.94 Hz of false spikes per neuron, versus 0.0321 Hz per spike. Under the per-bin model the more flexible decoders lose more: $\rho = 0.602$ (MLP), 0.542 (LSTM) and 0.547 (Transformer). They are the most accurate on clean data, but from SER 3×10^{-3} on Ridge gives the highest absolute r (0.774 vs 0.760 for MLP).

2. **Temporal integration.** Summing to 20-ms bins and smoothing the output raised Ridge retention at SER 10^{-2} from 0.758 to 0.840 (baseline re-estimated with new corruption draws), with no loss of clean accuracy ($r = 0.858 \rightarrow 0.873$; Fig. 2).

3. **Population size.** Retention did not rise with the number of units (SER 10^{-2}: 0.713 at $N = 20$, 0.748 at $N = 161$; Fig. 3), although clean accuracy did. The "population dilution" explanation of my previous report is not supported.

Reading against Lee et al. Under the per-bin model, the rate-based decoders here keep 94–96% at SER 10^{-3}, against 88.8% for their SNN. Data, channel count and decoder differ, so this gap is not a measured advantage.

What this changes. My previous report claimed a 30-point robustness advantage of rate-based decoders over the SNN. That comparison mixed two SER definitions and misdescribed the SNN, which itself bins spikes into 25-ms intervals [1]; it is withdrawn (Appendix A).

Limits. One session of one monkey; i.i.d. errors; no SNN or ASBIT channel simulation; 10-ms data cannot reproduce 1-ms effects exactly.

1 Motivation

In the ASBIT protocol each sensor sends a Gold-coded packet only when it detects an event, so link errors appear as missed or false spikes rather than corrupted samples [1]. The event error rate (EER) a link must reach therefore depends on how many spike errors the downstream decoder can absorb. Lee et al. found that network performance is set mainly by the total number of events per second (about 10^5 events/s at EER $< 10^{-3}$), and that the number of nodes is likely limited by wireless power transfer and SAR rather than bandwidth [1]. A decoder that tolerates more spike errors would relax the EER target. This report measures that tolerance for rate-based decoders and asks which property of the decoder explains it.

2 Methods

Data. CRCNS pmd-1, file `MM_S1_processed.mat` ("S1" denotes session 1, not somatosensory cortex) [2]. 161 units (67 M1, 94 PMd), 496 trials, 10-ms bins; 93.9% of unit-bins are empty, mean rate 6.41 Hz, 9.8 active units per bin.

Features and validation. For each trial separately, $\mathbf{x}_t = [\mathbf{s}_{t-19}, \ldots, \mathbf{s}_t]$ (20 bins $\times$ 161 units). Folds are split by trial; lag windows never cross trial boundaries. No spike-count transform and no output smoothing are used in the main analysis; smoothing ($\sigma = 3$ bins, applied within each trial) appears only in Sec. 3.3.

Decoders. Ridge (inputs standardised; penalty chosen by inner 3-fold trial-level CV from $\{1, 10, \ldots, 10^4\}$); MLP ($3220 \to 512 \to 256 \to 2$, dropout 0.2); 2-layer LSTM (hidden 128); 3-layer causal Transformer ($d = 128$, 4 heads). Neural networks use Adam (10^{-3}, weight decay 10^{-4}) with early stopping on 10% of training trials. Accuracy is Pearson r averaged over v_x and v_y.

Spike-error models. Decoders are trained once on clean data; only test trials are corrupted, with `N_CORRUPT_REPEATS` $= 5$ independent draws per fold and SER level.

- **Per spike** (previous report): SER $= (N_{\text{miss}} + N_{\text{false}})/N_{\text{spike}}$; each spike is deleted with probability $p/2$ and Binomial$(N_{\text{spike}}, p/2)$ false spikes are placed at random.
- **Per 1-ms bin** (my reading of Lee et al., who count a missed or false spike within a 1-ms bin as an error [1]; the denominator is not stated explicitly): each 10-ms bin is treated as ten 1-ms slots; a spike is missed with probability p, and each empty slot receives a false spike with probability p.

Under the per-bin model a neuron firing at λ Hz gains about $1000p$ false spikes per second; under the per-spike model it gains $\lambda p/2$. At the rates in this dataset the two differ by more than two orders of magnitude.

Statistics. $\rho = r_{\text{SER}}/r_{\text{clean}}$ per fold and draw; 95% bootstrap CIs over these values (not independent across draws within a fold; see Limitations). A drop is tested by a paired t-test on Fisher-z transformed per-fold correlations (the test Lee et al. used for their jitter comparison), with Benjamini–Hochberg correction across SER levels.

3 Results

3.1 Clean accuracy

	Ridge	MLP	LSTM	Transformer
r (mean v_x, v_y)	0.858	0.907	0.903	0.896
95% CI	[0.852, 0.865]	[0.902, 0.912]	[0.898, 0.910]	[0.890, 0.903]
r (v_x only)	0.858	0.909	0.904	0.899
r with smoothing $\sigma = 3$	0.869	0.922	0.909	0.905

Table 1: Clean decoding, trial-level 5-fold CV. Lee et al. report $r = 0.9324$ for v_x with an SNN on different M1 data; the numbers are not directly comparable.

3.2 Retention under the two SER definitions

	SER 10^{-3}		SER 10^{-2}		SER 10^{-1}	
Decoder	per spike	per 1 ms	per spike	per 1 ms	per spike	per 1 ms
Ridge	1.000	0.964	0.998	0.756	0.983	0.302
MLP	1.000	0.942	0.998	0.602	0.976	0.176
LSTM	1.000	0.948	0.998	0.542	0.978	0.080
Transformer	1.000	0.949	0.998	0.547	0.980	0.114
False spikes added (Hz)	0.00321	0.994	0.0321	9.94	0.321	99.4

Table 2: Retention ρ (mean over 5 folds $\times$ 5 draws). Full sweep, CIs and BH-corrected tests are in `results.json`.

Under the per-spike model, retention stays above 0.97 for every decoder up to SER 10^{-1}. Under the per-1-ms model it falls to 0.942–0.964 at 10^{-3} and to 0.542–0.756 at 10^{-2}; Ridge degrades least and LSTM most. Every per-1-ms drop from 10^{-4} upward survives Benjamini–Hochberg correction; per-spike drops become statistically detectable between 10^{-3} and 10^{-2}, depending on the decoder, but stay at or below 0.3% up to 10^{-2}. Ridge's penalty was chosen at the top of the grid ($\alpha = 10^4$, recorded for the last fold), so heavier shrinkage may raise its tolerance further.

3.3 Temporal integration

If binning acts as a low-pass filter on spike trains, wider bins should average out uncorrelated spike errors, whose power is spread across frequencies, while keeping the slow velocity-related signal. I re-binned the (already corrupted) 10-ms counts to 20 and 50 ms, keeping a 200-ms window, with and without output smoothing.

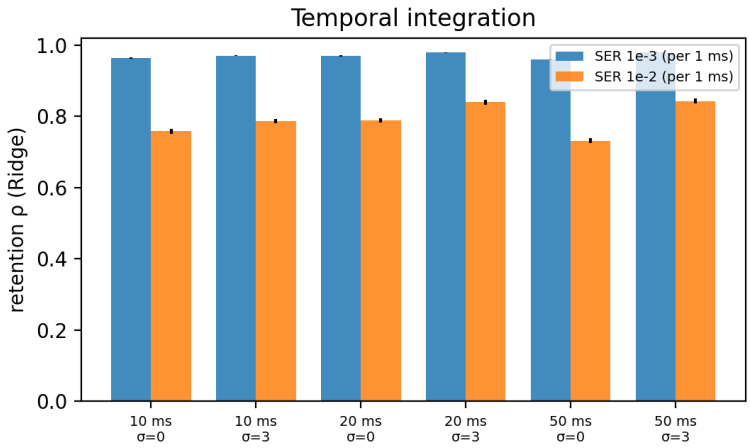


Figure 2: Ridge retention under per-1-ms errors for different bin widths and smoothing (σ in bins).

Going from 10- to 20-ms bins and adding output smoothing each raised retention at SER 10^{-2}; together, 20-ms bins with $\sigma = 3$ gave 0.840 [0.833, 0.846], against 0.758 for the 10-ms baseline. Clean accuracy was unchanged or higher (0.858, 0.873). At 50 ms without smoothing retention fell to 0.731, likely because only 4 lags remain; with smoothing it was 0.843 but clean r dropped to 0.844. Low-pass filtering helps, but recovers only part of the loss.

3.4 Population size and redundancy

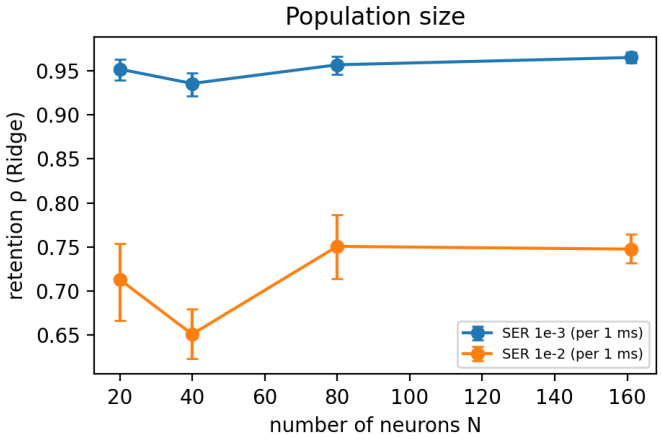


Figure 3: Ridge retention under per-1-ms errors for random subsets of N units (5 subsets each).

The participation ratio of smoothed population activity is 14.1 (out of 161 units), against 20.8 after shuffling time bins within trials. So population activity is low-dimensional relative to its size, and more so than after destroying temporal structure. Yet adding units did not improve retention (Fig. 3; CIs overlap from $N = 20$ to 161). One reading is that each added unit brings its own independent false spikes as well as signal, so redundancy improves clean accuracy without improving the ratio. Testing this needs a population-structure manipulation, not random subsampling.

4 Discussion

If the per-1-ms model is the right reading, rate-based decoders on this dataset need SER of about 10^{-3} or lower to keep roughly 95% of clean accuracy (Ridge 0.964; neural networks 0.942–0.949). This is close to the EER of about 10^{-3} that Lee et al. describe as acceptable for BMI use. Decoder-side

measures (wider bins, smoothing, heavier shrinkage) buy only part of a decade. This suggests, as a hypothesis, that most of the error budget will have to come from the link and its power delivery, rather than from choosing a more tolerant decoder.

A decoder's tolerance only matters relative to the error model of a real link. The per-bin model here assumes independent errors at a constant rate; Lee et al. show that a node's error rate depends mostly on its own backscatter amplitude, so near–far differences in SNR would make errors unequal across channels [1]. Testing decoders under that channel-dependent error pattern is the natural next step, and it connects decoder tolerance to the power and SAR budget that limits node count.

5 Limitations

- One monkey, one session; no cross-session test.
- The data are 10-ms bins, so 1-ms errors are emulated by sub-slot sampling.
- No SNN was trained and no ASBIT link was simulated; comparison with Lee et al. is qualitative only.
- The SER denominator in Lee et al. is my reading of the text and should be confirmed.
- Ridge's penalty hit the top of the search grid; a wider grid is needed.
- Confidence intervals pool fold × draw values (25 per condition); draws within a fold share one trained decoder and one test set, so they are not independent and the intervals are optimistic.

A Changes from the previous version (v2, June 2026)

- **Withdrawn: "30-point advantage over the SNN".** v2 compared retention under a per-spike SER with the SNN's retention, whose SER is (as I read it) counted per 1-ms bin, on different data. The per-spike model adds far fewer false spikes.
- **Corrected: the SNN does not use precise inter-spike intervals.** It bins 1-ms spikes into 25-ms intervals and is trained with ±25 ms jitter [1]; v2's timing-versus-rate explanation was wrong.
- **Corrected attribution.** The ASBIT paper is Lee, Lee et al. (2024), not "Nurmikko et al."; ASBIT is the network protocol, not the decoder.
- **Removed unsupported claims**: typical link BER values and power savings attributed to the paper.
- **Analysis**: dropped the spike-count exponent α (chosen on a test fold in v2); removed default output smoothing; corruption repeated 5 times instead of once; added CIs and BH correction.
- **Dataset**: pmd-1 is from Perich, Lawlor, Kording and Miller and involves sequential reaching [2]; v2 cited it incorrectly.
- **Attention analysis** moved out of the main text; the attention weights in v2 (about 0.06–0.07 per bin against a uniform 0.05) do not support a strong recency claim.

References

[1] J. Lee, A.-H. Lee, V. Leung, F. Laiwalla, M. A. Lopez-Gordo, L. Larson and A. Nurmikko, "An asynchronous wireless network for capturing event-driven data from large populations of autonomous sensors," *Nature Electronics* **7**, 313–324 (2024). doi:10.1038/s41928-024-01134-y

[2] M. G. Perich, P. N. Lawlor, K. P. Kording and L. E. Miller, "Extracellular neural recordings from macaque primary and dorsal premotor motor cortex during a sequential reaching task," CRCNS.org (pmd-1). doi:10.6080/K0FT8J72